# Persistent K-Coverage: Where the Progress Really Stands

Professor progress meeting — 4 November 2026

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# Agenda (15 minutes)

1. What the report is now
2. What we fixed, and what we hid nowhere
3. The measured headline: where learning wins and loses
4. The energy-flow certificate
5. Planning oracle and the deployable distilled policy
6. Honest verdict, risks, and what we need from you

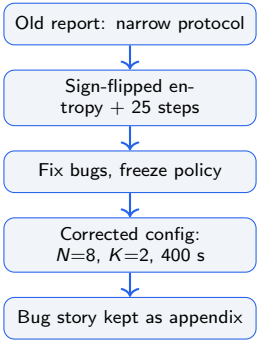
# What the report is now

- 27 pages; title changed to *"Energy-Flow Feasibility and Learned Redundancy Management for Persistent K-Coverage UAV Networks"*.
- Story order: certificate $\rightarrow$ controller $\rightarrow$ honest negatives.
- Frozen policy; held-out scenes (seeds $\geq 1000$); paired 95% intervals.
- Bug story moved to the appendix; the headline number is the corrected one.

**The evaluated observation is 68-dimensional**

`30 base + 16 map grid + 8 agent ID + 3 pad bearing + 11 ego-centric deficit bearings`

# What broke, and what fixed it



# Measured headline (report)

| Comparison | CK | Notes |
|---|---|---|
| MAPPO (ego + shaping) | 0.9512 | held-out, 20 scenes |
| Strongest rule (Reactive-local) | 0.923 | +0.028, CI excludes zero |
| Oracle ceiling | 1.000 | oracle placement, identical for all |
| Classical OR scheduler | 0.929 | oracle + Hungarian + rotation |

- Learning is ahead on nominal and low connectivity; behind on small battery and spread start.
- Bug-fixes-only: 0.859; anchor ego+shaping: 0.9512.

# Stress regimes (5 held-out scenes)

| Regime | Best rule | MAPPO | Classical OR | MAPPO − rule |
|---|---|---|---|---|
| nominal | 0.921 | 0.947 | 0.929 | +0.026 |
| small battery | 0.829 | 0.765 | 0.791 | −0.064 |
| low comm | 0.804 | 0.928 | 0.808 | +0.125 |
| spread start | 0.997 | 0.959 | 0.980 | −0.038 |
| failure | 0.912 | 0.946 | 0.919 | +0.034 |
| multi-failure | 0.910 | 0.945 | 0.919 | +0.035 |

- Classical scheduler does not beat the learner in any same-architecture regime.
- The strongest win is low connectivity: 0.928 vs 0.808 (classical) and 0.804 (rule).

# Energy-flow certificate (new)

- Each of the 2 pads adds 30 W gross, 24.9 W net after avionics.
- One airborne UAV draws 55.1 W.
- Steady state: only 0.90 UAV can remain airborne.
- $K$=2 needs 8 UAVs.
- Steady-state ceiling = 0.113.
- $T$=400s: ceiling 1.000 (spends initial packs).
- $T$=2000s: ceiling 0.367 (full) / 0.326 (mean charge).
- Only $N$=12, 4 pads, 60 kJ, fast charge achieves 1.000.

**The reported configuration is not long-horizon persistent.** That is the paper's own, verified finding.

# Planning oracle and the deployable policy

- Centralised planning oracle: 0.9723 vs base 0.9512. Paired $+0.0211$, CI $[+0.014, +0.028]$, resolved.
- It is global, not deployable CTDE.
- Distilled 20-epoch actor: 0.9618, paired $+0.0106$, CI $[-0.002, +0.023]$ (directional).
- Planner recoverable as deployable decentralised policy: about half the gain.
- Classical OR scheduler: 0.929, still below the learner.

# Honest verdict

- Not a breakthrough; a correct, honest engineering package.
- The two "certificates" are conservation, not new theory.
- The value is the certificate, corrected protocol, and honest negatives.
- Realistic venue: NCC / COMSNETS / VTC-ICC workshop / Drones.
- Not top-tier (TMC / NeurIPS / ICRA).

The energy certificate makes the paper coherent, not novel in the strongest sense.

# Open issues (we are showing before hiding them)

1. **Observation documentation bug:** the report text describes the observation incorrectly; the evaluated 68-dim model includes map grid, agent ID, pad bearing and ego features.
2. **Not long-horizon persistent:** the current 400 s scope must stay explicit.
3. **Title tension:** energy-first title may invite a rejection before the evidence is read.
4. **0.862 "full coverage" claim:** provenance and architecture not yet audited.

# What we need from you

1. Venue scope: regional conference, Drones, or hold until the energy / low-comm story is cleaner.
2. Title: keep energy-first wording, or move certificate contribution into the abstract and lead with the system.
3. Scope: a) fix documentation and submit as-is; b) run 2 more experiments ($R_{\mathrm{comm}}$ phase transition; proactive staggered recharging); c) pivot paper to $\mathrm{low}_{\mathrm{comm}}$.

# Next two weeks (proposal)

1. Fix the observation/config documentation and provenance of 0.862.
2. $R_{\mathrm{comm}}$ phase-transition sweep with no retraining.
3. Decide venue and title.
4. Final submission package: 10 pages + appendix/info-pack.

# Bottom line

**We have a real, corrected, honest contribution, not a breakthrough.**

- Certificate tells us when the task is feasible, including energy.
- The learner beats the strongest rule on nominal and low connectivity.
- It loses where energy is the constraint; that is the paper's point.
- The report is in much better shape, but it still needs one documentation fix before submission.